Online Supplement

Grandmothers' care and mothers' time: coresident grandmothers in India

Journal of Population Economics

Preet Deep Singh, Independent Researcher, New Delhi, India

Corresponding author: preetds@iima.ac.in

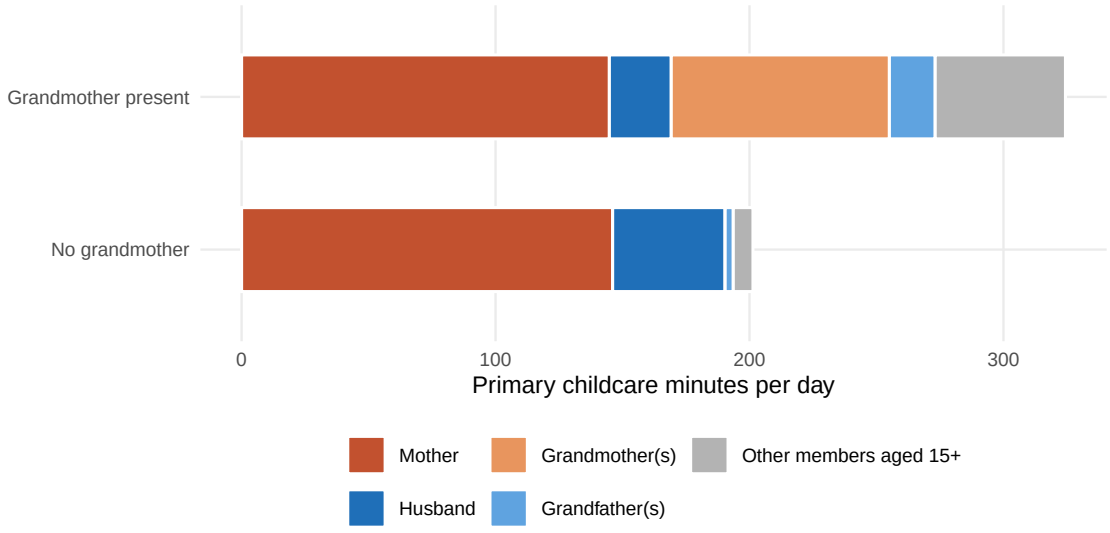

Figure S1: Primary childcare minutes per day by provider, with and without a coresident grandmother. Values and standard errors in Table 3.

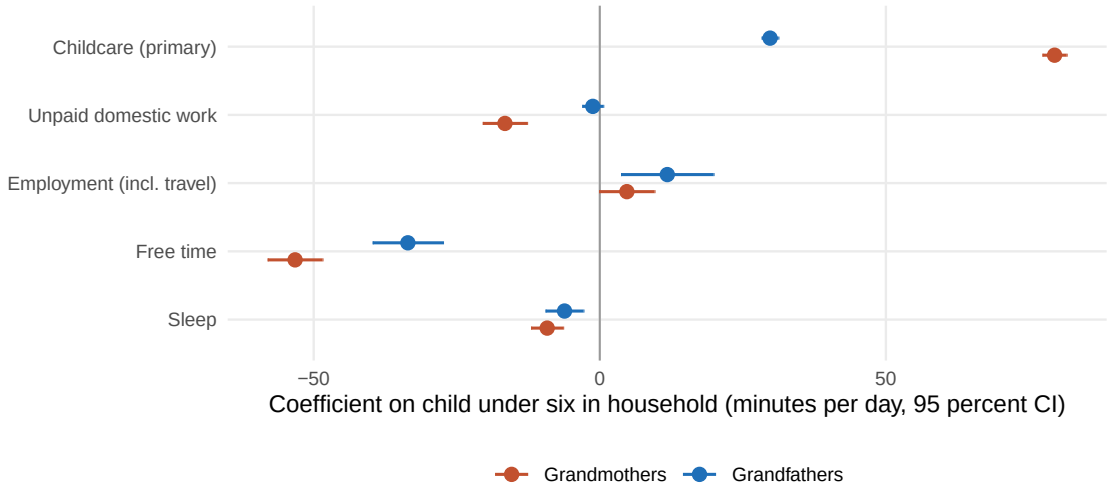

Figure S2: Older generation: minutes per day associated with a child under six in the household. Estimates in Table 4.

Table S1: Construction and validation of the person-day panel

Check	Value
Activity records	10,211,478
Person-days (diaries)	450,457
Households	139,487
First Stage Units	9,969
Diaries not summing to 1,440 minutes	5 of 450,457
Records flagged minor (secondary)	17.5%
Persons with varying weight	0.00%
Invalid activity codes	None
Married persons with linked spouse	88.5%
Under-6 count method	direct count (file contains members under 6)

Notes: All activity records are used; no row sampling. The number of children under six is a direct count of roster members under six.

Table S2: Sample construction

Step	N
Married women 18–49 at head or junior level, child under six in household	52,181
Excluded: head-level women living with a junior married woman aged 15–49	2,189
Analysis sample (mothers of children under six)	49,992
with a coresident grandmother	18,028
junior-level mothers without a grandmother	2,054
head-level mothers without a grandmother	29,910
junior-level mothers with a grandmother	15,196
head-level mothers with a grandmother	2,832
Placebo sample (youngest own child 6–14, no child under six)	19,829
Older generation: women aged 45–80	26,062
Older generation: men aged 45–80	19,382
Weighted shares, analysis sample	
with a coresident grandmother	36%
with a coresident grandfather	28%
with a grandfather but no grandmother	4.5%
at junior level	33%
daughters-in-law of the head, among junior-level mothers	95%
with a linked husband	83%
at junior level, among mothers with a grandmother	83%
youngest grandmother under 60, among mothers with a grandmother	60%
grandmother employed (usual status), among mothers with a grandmother	30%
grandmother(s)' share of mother's plus grandmother(s)' childcare	37%
Median age of the youngest coresident grandmother (years)	57
Mothers' employment by generation and grandmother (share employed; employment minutes a day)	
head-level, no grandmother	0.178; 48.8
head-level, with a grandmother	0.227; 67.3
junior-level, no grandmother	0.142; 37.3
junior-level, with a grandmother	0.153; 46.1

Notes: Unweighted counts of person-day diaries, then survey-weighted shares; the median age is unweighted; the employment rows are survey-weighted means (usual-status employed share; minutes of employment including related travel). Head level: head or spouse of head. Junior level: married child of head or spouse of a married child.

Table S3: Older-generation women: sensitivity to the lower age limit

Outcome	Women 45–80 (main)	Women 40–80	Women 50–80
Childcare (primary)	79.48 [77.37, 81.60]	79.14 [77.06, 81.22]	79.00 [76.78, 81.22]
Active childcare	70.43 [68.40, 72.45]	70.08 [68.09, 72.08]	69.72 [67.60, 71.84]
Passive childcare (minding)	7.65 [6.82, 8.47]	7.66 [6.84, 8.47]	7.89 [7.01, 8.78]
Unpaid domestic work	-16.58 [-20.45, -12.70]	-16.19 [-19.99, -12.38]	-14.13 [-18.14, -10.11]
Employment (incl. travel)	4.73 [-0.11, 9.57]	3.88 [-0.94, 8.70]	5.24 [0.48, 10.01]
Employed (usual status, share)	0.016 [0.004, 0.028]	0.015 [0.003, 0.027]	0.017 [0.004, 0.029]
Free time	-53.25 [-58.04, -48.46]	-52.58 [-57.28, -47.88]	-54.63 [-59.70, -49.56]
Sleep	-9.19 [-11.99, -6.39]	-9.05 [-11.79, -6.31]	-10.33 [-13.29, -7.37]

Notes: Estimate [95 percent confidence interval] for a child under six in the household, specification as in the older-generation table. Main sample: 26,062 women aged 45–80; alternative: 26,986 women aged 40–80, the same lower limit as the grandmother link for junior-level mothers; and 23,228 women aged 50–80, which drops the 2,834 women aged 45–49 who could be the mothers of a late-born child. Conversely, 544 mothers in the analysis sample have a coresident grandmother only through a woman aged 40–44; in the robustness-table rows that define the grandmother as aged 45 or above, these mothers stay in the sample and are counted as having no coresident grandmother.

Table S4: Components of unpaid domestic work (minutes a day)

Component	Mothers: grandmother present	Older women: child under six present
Food and meals	5.01 [2.34, 7.67]	-10.47 [-13.41, -7.52]
Cleaning	-4.50 [-5.57, -3.43]	-2.62 [-3.88, -1.36]
Care of textiles	0.38 [-0.78, 1.53]	-1.65 [-2.60, -0.71]
Shopping	-0.73 [-1.12, -0.33]	-0.37 [-0.92, 0.19]
Household management	0.79 [0.30, 1.28]	-0.18 [-0.80, 0.44]
Repair and maintenance	0.10 [-0.09, 0.28]	0.09 [-0.11, 0.30]
Pet care	-0.11 [-0.29, 0.07]	-0.05 [-0.42, 0.32]
Travel for domestic work	-0.04 [-0.27, 0.18]	-0.28 [-0.59, 0.02]
Other domestic work	0.25 [-0.13, 0.63]	-1.05 [-1.71, -0.39]

Notes: Estimate [95 percent confidence interval]. Column 1: coefficient on the grandmother indicator in the main specification for mothers (49,992). Column 2: coefficient on a child under six in the household for older-generation women (26,062), specification as in the older-generation table.